

rated for years, often generating more heat than light. Many alarming claims require careful examination of their origins and supporting evidence.

Security researchers have documented legitimate vulnerabilities in both systems. Intel's ME has experienced multiple security flaws over the years, including vulnerabilities that could potentially allow unauthorised remote access. However, according to Intel's documentation, every documented remote code execution vulnerability in Intel's ME

One documented PSP vulnerability (CVE-2021-26333) affected the firmware TPM functionality and could potentially allow attackers with low-level system access to obtain sensitive information. However, this vulnerability required direct system access and was promptly patched when discovered.

The absence of definitive proof supporting deliberate backdoor claims doesn't mean these systems are without security concerns. The proprietary nature of both Intel's ME and AMD's PSP firmware means that inde-

have requested such an option from us, and in response, we have provided the service of disabling the Management Engine in the factory to meet their specific needs". This commercial availability suggests that government agencies and security-conscious organisations have indeed sought ways to operate without these embedded processors.

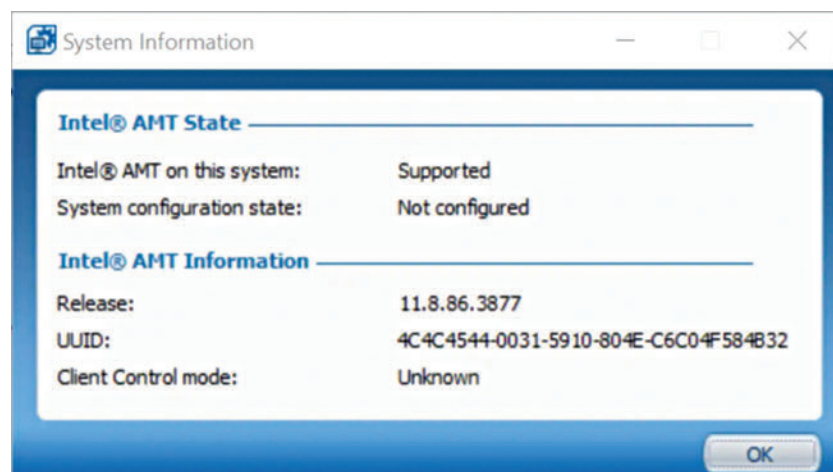
The geopolitical dimensions have intensified recently, with China's Cyber-security Association formally accusing Intel of embedding backdoors in its processors at the direction of the NSA. Intel has responded by stating it "strictly abides by the laws and regulations applicable to its business locations", neither confirming nor denying the specific technical claims.

Security vulnerabilities in practice

Both Intel ME and AMD PSP have experienced documented security vulnerabilities that demonstrate the real-world risks these systems can pose.

Intel's ME has suffered from multiple vulnerabilities over the years. These flaws have ranged from privilege escalation bugs that could allow attackers to gain elevated system access to remote code execution vulnerabilities that could allow system compromise. Intel has responded to these discoveries by releasing firmware updates and security patches, though the update process often requires specific actions from computer manufacturers or end users.

AMD's PSP has experienced fewer documented vulnerabilities, partly due to its limited network exposure. The most significant PSP vulnerability (CVE-2021-26333) affected the firmware TPM implementation and was discovered in 2018. This flaw could potentially allow attackers with appropriate system access to extract information from the TPM, though it requires physical or local access to exploit. **d**



has required Active Management Technology to be explicitly enabled and provisioned.

This distinction matters considerably. AMT represents an intentional remote access feature that organisations must deliberately configure. If you enable remote access capabilities on any system, vulnerabilities in those features create potential security risks. The pattern suggests that Intel's ME functions more like any other remote management system, with similar associated risks, rather than a deliberately planted backdoor.

AMD's PSP presents a different security profile entirely. The lack of network access means that "only one with physical access could exploit" any potential vulnerabilities. This significantly reduces the scope for remote attacks, though it doesn't eliminate security concerns entirely.

pendent security researchers cannot conduct thorough audits of their code. This opacity naturally breeds suspicion and makes it impossible to definitively rule out intentional vulnerabilities.

Government involvement

The most credible evidence of government involvement emerged from a hidden firmware bit in Intel's ME called "reserve_hap" (High Assurance Platform enable). This hidden feature can disable the ME after system boot-up, and security experts believe Intel added this capability specifically at the request of the NSA for use in highly sensitive government environments.

Dell has openly offered commercial customers the option to purchase systems with Intel ME disabled, marketing this service as "Intel vPro – ME inoperable" for custom orders. Dell stated that "some of our commercial customers

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